

Java Darleen Villano

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Academic positions

University of Toronto <i>Postdoctoral Fellow</i>	2025-2026
University of Connecticut <i>Graduate Student</i>	2019-2025

Education

University of Connecticut <i>Ph.D. Mathematics</i>	2019-2025
◦ Advisers: Reed Solomon and Damir D. Dzhafarov	
◦ Dissertation Title: Computable Categoricity, and Topology in Reverse Mathematics	
University of California, Berkeley <i>B.A. Mathematics with Logic Minor</i>	2015-2019

Research interests

Computability theory, computable structure theory, reverse mathematics, Weihrauch complexity, and algorithmic randomness.

Publications

Extensions of categoricity relative to a degree	June 2026
Villano, J.D.	
Accepted, <i>Zeitschrift für Mathematische Logik und Grundlagen der Mathematik</i> 🔗	
Computable categoricity relative to a c.e. degree	November 2025
Villano, J.D.	
<i>Notre Dame Journal of Formal Logic</i> 🔗	
Normality, relativization, and randomness	June 2025
Calvert, W., Gruner, E., Mayordomo, E., Turetsky, D., Villano, J.D.	
<i>Theory of Computing Systems</i> 🔗	
The Ginsburg–Sands theorem and computability theory	May 2024
Benham, H., DeLapo, A., Dzhafarov, D., Solomon, R., Villano, J.D.	
<i>Advances in Mathematics</i> 🔗	

Preprints

Switching Categoricity Behavior with a Given Degree	September 2026
Gonzalez, D., Valenzuela Morales, J.J., Villano, J.D.	
arXiv:2609.21047 🔗	

Teaching experience

Primary Instructor	Toronto, ON
<i>University of Toronto</i>	2025-2026
◦ Fall 2025, Winter 2026: MAT133Y (Calculus and Linear Algebra for Commerce), 1 section	
Primary Instructor	Storrs, CT
<i>University of Connecticut</i>	2023-2024
◦ Fall 2024: Math 1071Q (Calculus for Business and Economics), 2 sections	
◦ Spring 2024: Math 1071Q (Calculus for Business and Economics), 1 section	
◦ Fall 2023: Math 1071Q (Calculus for Business and Economics), 2 sections	

- **Spring 2023:** Math 1071Q (Calculus for Business and Economics), 2 sections

Teacher Assistant

University of Connecticut

*Storrs, CT
2019-2022, 2025*

- **Spring 2025:** Math 2110Q (Multivariable Calculus), 3 sections
- **Fall 2022:** Math 1132Q (Calculus II), 2 sections
- **Spring 2022:** Math 1132Q (Calculus II), 2 sections
- **Fall 2021:** Math 1131Q (Calculus I), 2 sections
- **Spring 2021:** Math 1132Q (Calculus II), 2 sections
- **Fall 2020:** Math 1132Q (Calculus II), 2 sections
- **Spring 2020:** Math 1132Q (Calculus II), 2 sections
- **Fall 2019:** Math 1131Q (Calculus I), 2 sections

Conference invitations

AMS Fall Central Sectional Meeting

St. Louis University

*St. Louis, MO
Fall 2025*

Presentation title: Categorical behavior in the Turing degrees

Workshop “Reverse Mathematics: New Paradigms”

Erwin Schrödinger International Institute for Mathematics and Physics

*Vienna, Austria
Summer 2025*

Presentation title: Relativized computable categoricity

Summer School “Reverse Mathematics: New Paradigms”

Erwin Schrödinger International Institute for Mathematics and Physics

*Vienna, Austria
Summer 2025*

Logicón 2025

Facultad de Ciencias UNAM

*México City, México
Spring 2025*

Presentation title: Computable categoricity relative to a degree (online talk)

ASL North American Annual Meeting

New Mexico State University

*Las Cruces, NM
Spring 2025*

Presentation title: Computable categoricity relative to a generic degree

Dagstuhl Seminar – Weihrauch Complexity: Structuring the Realm of Non-Computability

Schloss Dagstuhl

*Wadern, Germany
Spring 2025*

South Eastern Logic Symposium

University of Florida

*Gainesville, FL
Spring 2025*

Presentation title: Computable categoricity relative to a degree

Graduate Research Forum

University of Connecticut

*Storrs, CT
Spring 2025*

Presentation title: Relativizing computable categoricity

The New England Recursion and Definability Seminar

Dartmouth College

*Hanover, NH
Fall 2024*

Presentation title: Computable categoricity relative to a c.e. degree

Computable Structure Theory and Interactions

Technische Universität Wien

*Vienna, Austria
Summer 2024*

Presentation title: Computable categoricity relative to a degree

Joint Mathematics Meeting – AMS Special Session on Computable Mathematics: A Session Dedicated to Martin D. Davis

Presentation title: Computable categoricity relative to a c.e. degree

*San Francisco, CA
Spring 2024*

A Convergence of Computable Structure Theory, Analysis, and Randomness

Banff International Research Station

*Banff, AB
Spring 2023*

AMS New England Graduate Student Conference
Brown University
Presentation title: Priority arguments

Providence, RI
Spring 2022

Contributed presentations

ASL North American Annual Meeting
Iowa State University

Ames, IA
Spring 2024

Presentation title: Computable categoricity relative to a c.e. degree

AMS New England Graduate Student Conference
Brown University

Providence, RI
Spring 2024

Presentation titles: Topology in the Reverse Math Zoo; Computable categoricity relative to a c.e. degree

17th International Conference on Computability, Complexity, and Randomness
Nagoya University

Nagoya, Japan
Spring 2024

Presentation title: Computable categoricity relative to a c.e. degree

Conferences and workshops attended

CoCoGEMS @TU Wien
Technische Universität Wien

Vienna, Austria
Summer 2026

CBMS Conference – Algorithmic Fractal Dimensions
Drake University

Des Moines, IA
Spring 2024

Computability and Combinatorics Summer School and Conference
UConn Hartford

Hartford, CT
Spring 2023

ASL Winter Meeting at the Joint Mathematics Meeting

Boston, MA
Spring 2023

IMS Graduate Summer School in Logic
National University of Singapore

Singapore
Summer 2022

Seminar presentations

Logic Seminar
George Washington University

Washington, D.C.
Spring 2026

Presentation title: Relativizing categoricity and low c.e. sets

Logic Seminar
University of Waterloo

Waterloo, ON
Fall 2025

Presentation title: Relativizing computable categoricity

SIGMA Seminar
University of Connecticut

Storrs, CT
Spring 2025

Presentation title: The Scott Isomorphism Theorem

Online Logic Seminar
Southern Illinois University

Online
Fall 2024

Presentation title: Computable categoricity relative to a degree

SIGMA Seminar
University of Connecticut

Storrs, CT
Spring 2024

Presentation title: The Ginsburg–Sands theorem and computability theory

SIGMA Seminar
University of Connecticut

Storrs, CT
Spring 2024

Presentation title: Normality and Randomness

SIGMA Seminar
University of Connecticut

Storrs, CT
Fall 2023

Presentation title: Randomness and Hausdorff dimension

Connecticut Logic Seminar*University of Connecticut**Storrs, CT**Fall 2023***Presentation title:** Computable categoricity relative to a c.e. degree**SIGMA Seminar***University of Connecticut**Storrs, CT**Fall 2022***Presentation title:** When does the existence of an isomorphism imply the existence of a computable isomorphism?**Grants and funding**

Summer Doctoral Dissertation Fellowship*\$2,000 USD**Summer 2024***Predoctoral Fellowship***\$7,805 USD**Spring 2024***Outreach**

President of the Association of Women in Mathematics*University of Connecticut**Storrs, CT**2022-2024***Speaker at the Mathematics Continued Conference***University of Connecticut**Storrs, CT**Fall 2022*

The Mathematics Continued Conference seeks to give undergraduate students interested in math an opportunity to learn about graduate school and current research done by graduate students and faculty.

Course Tutor for SSS Math Program*University of Connecticut**Storrs, CT**Summer 2020*

Student Support Services (SSS) is a federally funded program at UConn which serves incoming students who are first-generation to college and/or come from communities underserved in higher education.

Languages

English*Advanced proficiency in reading, writing, and speaking**Second language learned,**learned in 2003***Tagalog***Intermediate proficiency in reading, writing, and speaking**Native language*